

Quantification of atmospheric carbon dioxide from the Geostationary Operational Environmental Satellite (GOES East)

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Abstract

There is a growing urgency to track greenhouse gasses with the resolution, precision and accuracy needed to support independent verification of CO₂ fluxes at local to global scales. The current generation of space-based sensors, however, only provides sparse observations in space and time. This challenge has fueled interest in the potential use of data from existing missions originally developed for other applications for inferring global greenhouse gas variability. The Advanced Baseline Imager (ABI) onboard the Geostationary Operational Environmental Satellite (GOES-East), operational since 2017, provides full coverage of much of the western hemisphere at 10-minute intervals from geostationary orbit at 16 wavelengths at an approximately 2km² spatial resolution. Here, we leverage this high spatial coverage and temporal revisit to develop a single-pixel, physics-guided neural network to estimate dry-air column CO₂ mole fraction (XCO₂). The model employs a time series of GOES-East's 16 spectral bands, ECMWF ERA5 lower tropospheric meteorology, MODIS surface reflectance, solar angles, and day of year. Training used collocated GOES-East and OCO-2/OCO-3 observations. The model is able to capture realistic XCO₂ variability when compared against a held-out year of OCO-2 and OCO-3 observations, and against observations from the TCCON network. We also present case studies illustrating the use of the model to observe XCO₂ enhancements over urban areas and drawdown over agricultural regions. Overall, while the precision of GOES-East derived XCO₂ can never rival that of dedicated instruments, the unprecedented combination of contiguous geographic coverage, 10-minute temporal frequency, and multi-year record offers the potential to observe aspects of atmospheric CO₂ variability currently unseen from space.